



GPU-Accelerated Ray Tracing of Catmull-Clark Subdivision Surfaces with OptiX using a Grid-Layout

Ray Tracing of Subdivision Surfaces with OptiX

| Agenda

- | | | | |
|----------|------------------------------------|----------|-----------------------|
| 1 | GPU Ray Tracing with OptiX | 4 | OptiX Implementations |
| 2 | Catmull-Clark Subdivision Surfaces | 5 | Evaluation |
| 3 | Patches and Grid-Layout | 6 | Outlook |

1 GPU Ray Tracing with OptiX

Ray Tracing



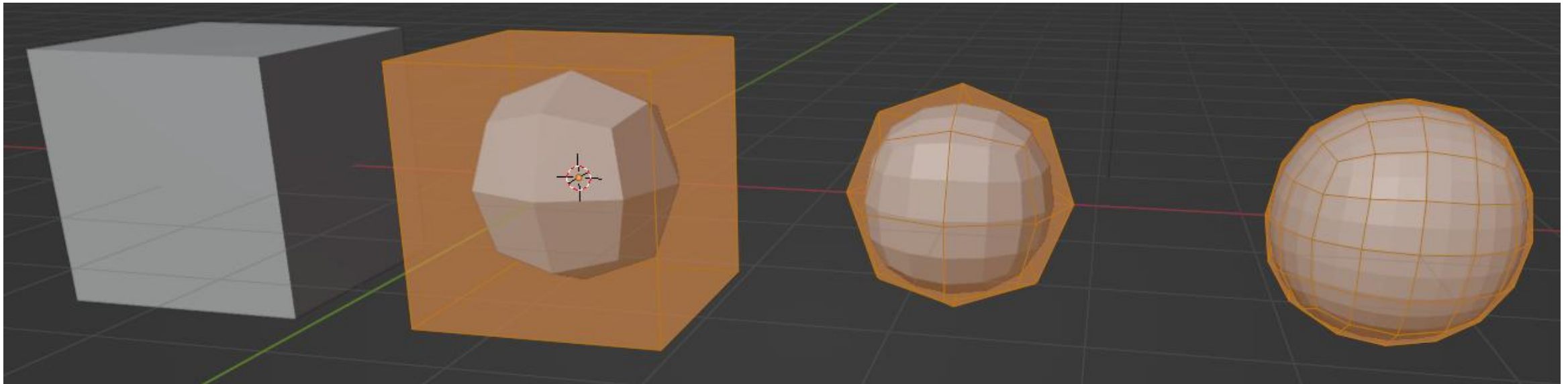
1 GPU Ray Tracing with OptiX

| NVIDIA OptiX



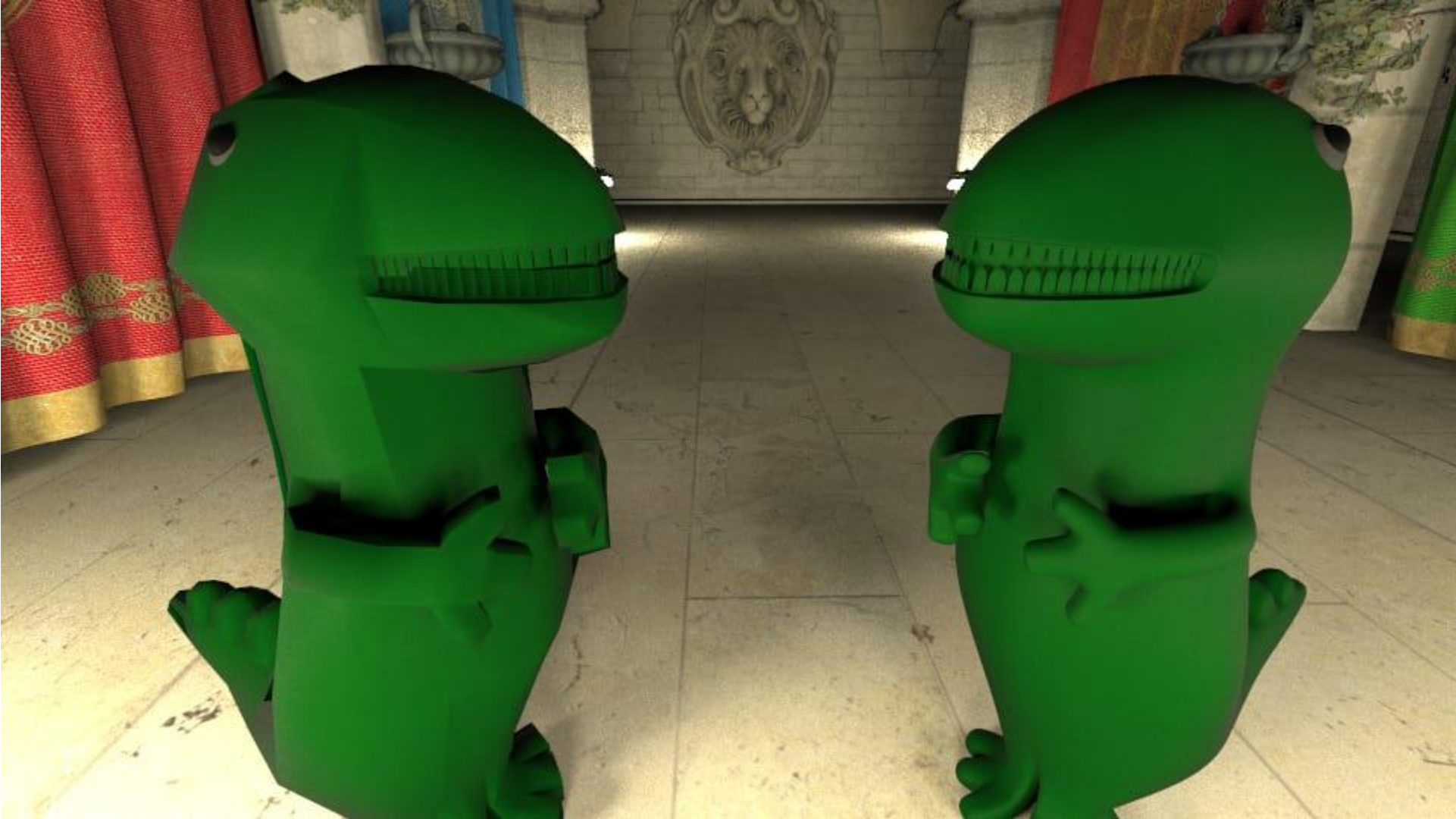
2 Catmull-Clark Subdivision Surfaces

Tessellation



2 Catmull-Clark Subdivision Surfaces

Comparison Base and Subdivided Mesh



2 Catmull-Clark Subdivision Surfaces

Comparison Base and Subdivided Mesh



2 Catmull-Clark Subdivision Surfaces

Comparison Base and Subdivided Mesh

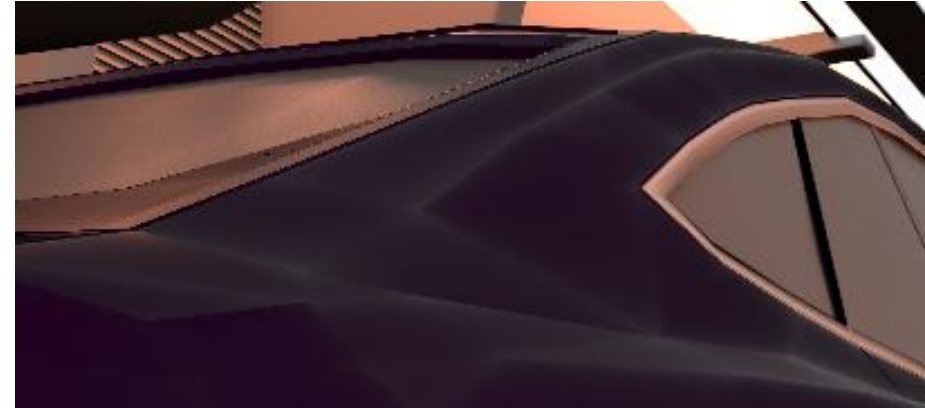


2 Catmull-Clark Subdivision Surfaces

Comparison Base and Subdivided Mesh



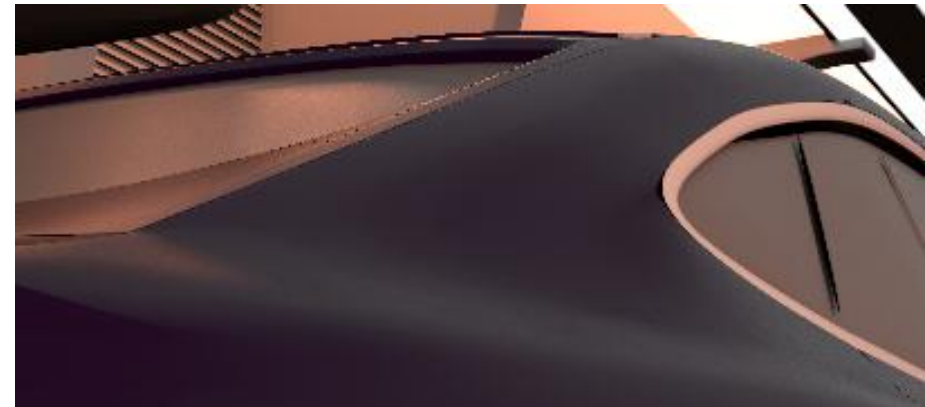
Base Mesh – Car Light



Base Mesh – Lighting



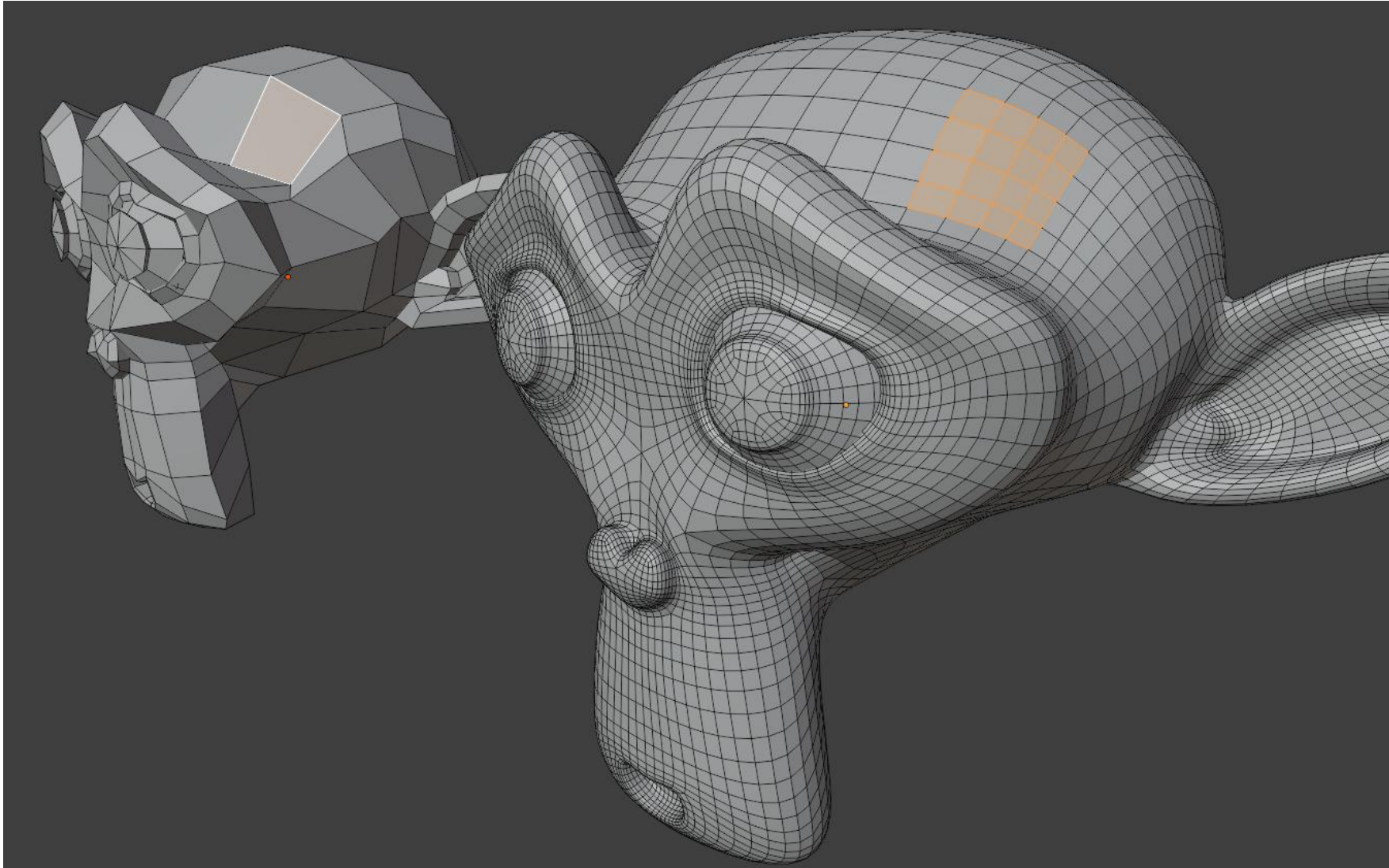
Subdivision Level 4 – Car Light



Subdivision Level 4 – Lighting

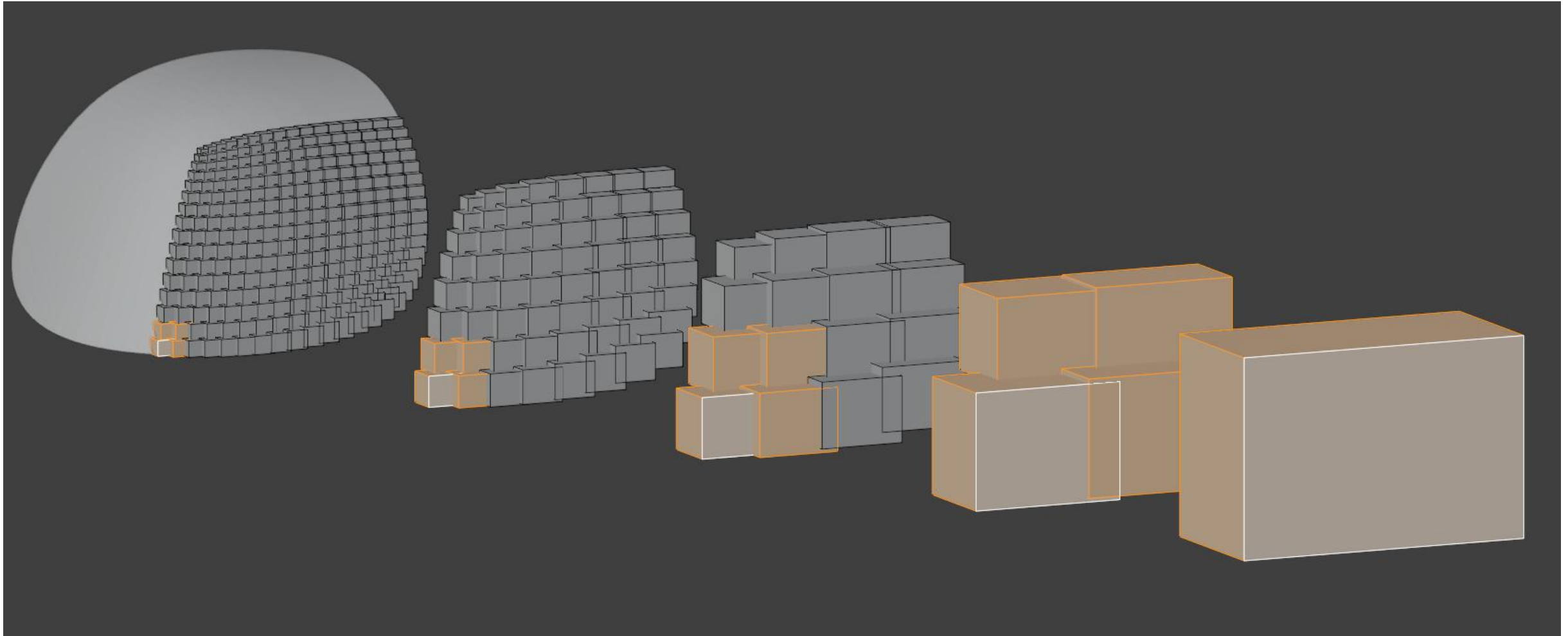
3 Patches and Grid-Layout

Structures



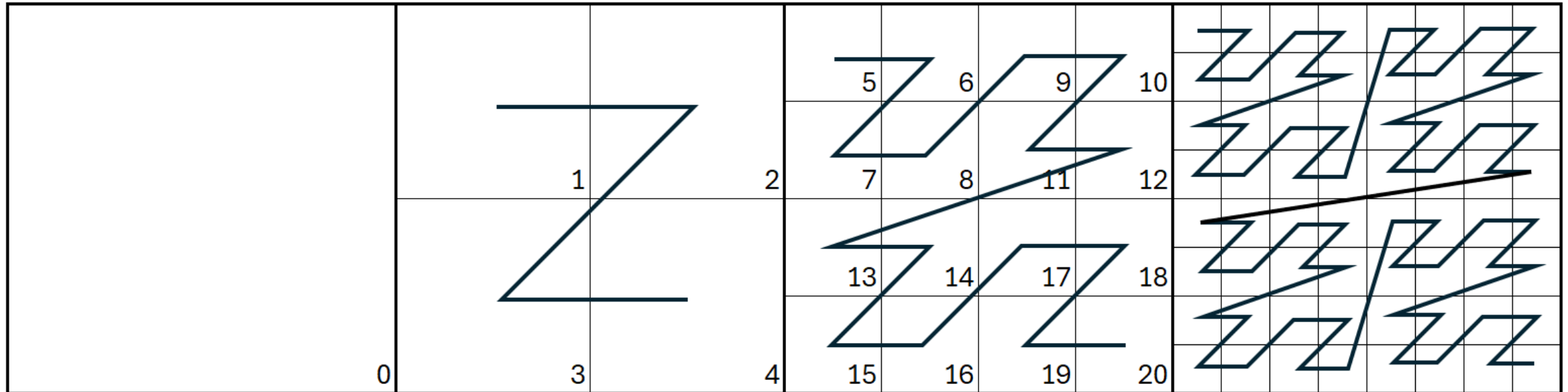
3 Patches and Grid-Layout

| Bounding Volume Hierarchy



3 Patches and Grid-Layout

Memory Alignment and Implicit Indexing



3 Patches and Grid-Layout

1 Block = 4 Bytes

Compact Node Layout

Simple Node		
		Blocks
float4	min	4
float4	max	4
uint4	child_refs	4
Total		12

→ Represents only one Bounding Box

Size four simple Nodes:
48 Blocks (192 Bytes)

Size of one Compact Node:
24 Blocks (96 Bytes)

Compact Node		
		Blocks
float4	data_1	4
float4	data_2	4
float4	data_3	4
float4	data_4	4
float4	data_5	4
float4	data_6	4
Total		24

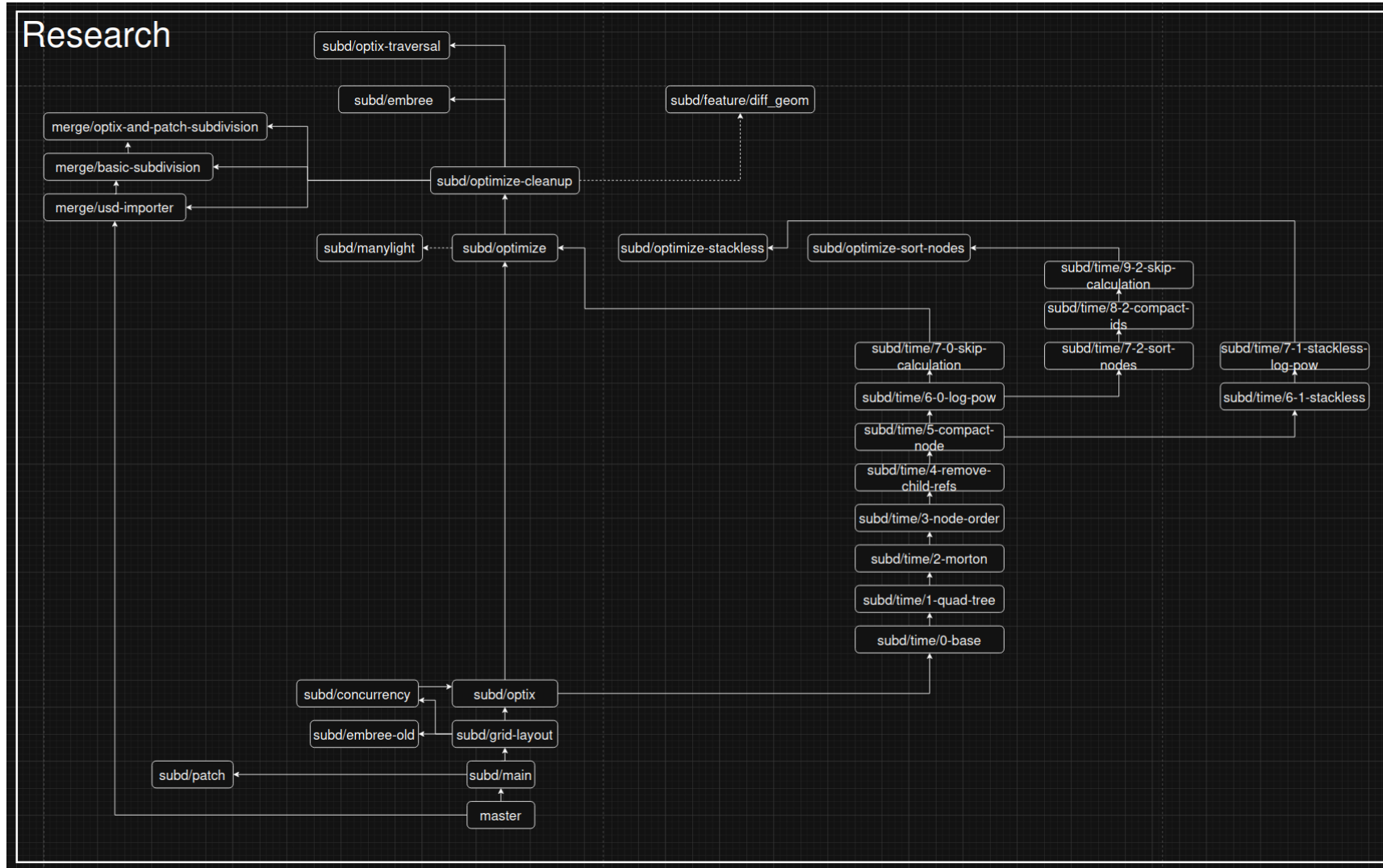
→ Represents four Bounding Boxes

Blocks

4	x_min	y_min	z_min	x_max
8	y_max	z_max	x_min	y_min
12	z_min	x_max	y_max	z_max
16	x_min	y_min	z_min	x_max
20	y_max	z_max	x_min	y_min
24	z_min	x_max	y_max	z_max

4 OptiX Implementations

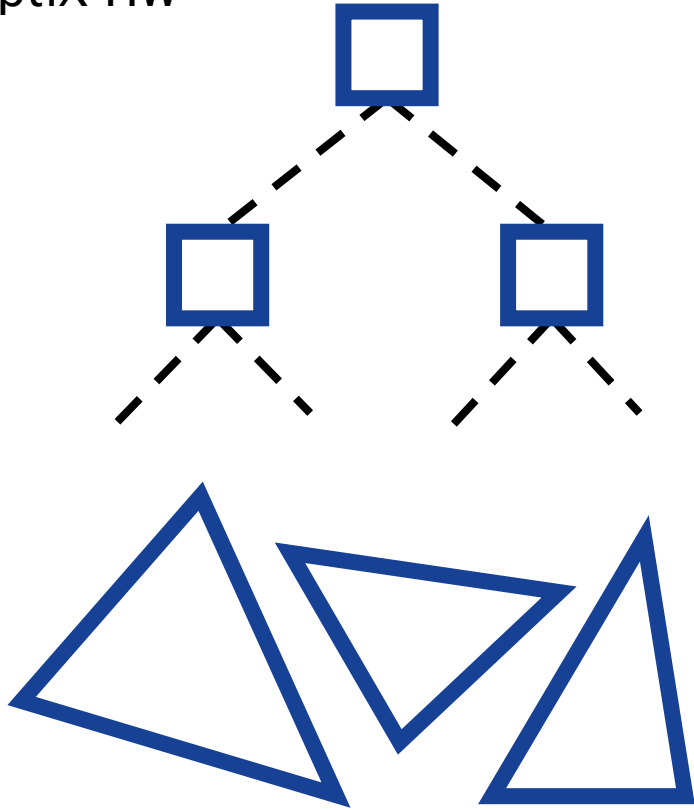
Git Branches



4 OptiX Implementations

Main Implementations

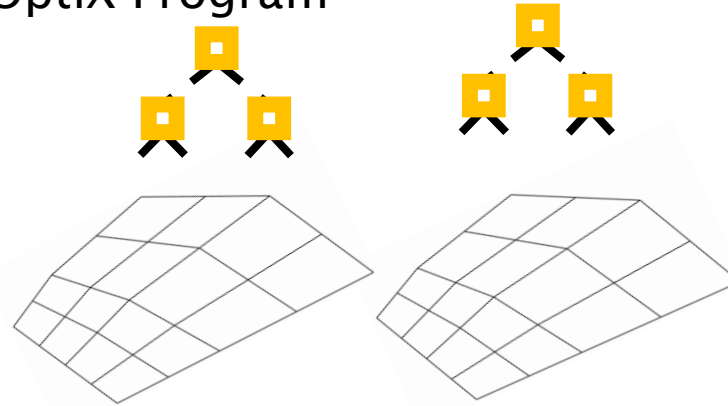
OptiX HW



Basic Triangles

OptiX RT HW

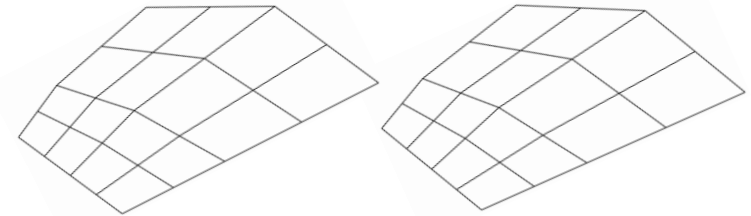
OptiX Program



Full Patch Traversal

OptiX RT HW

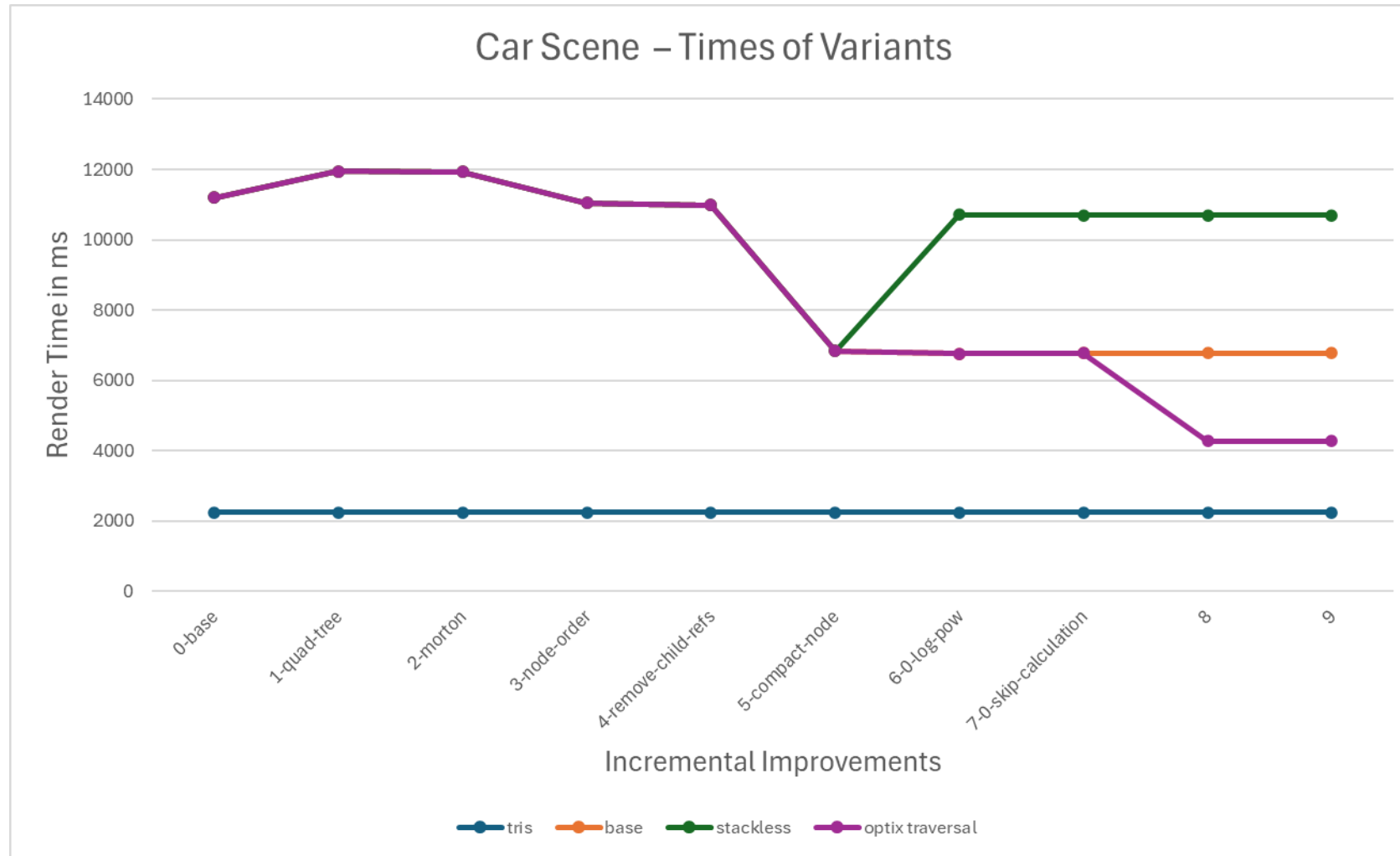
OptiX Program



Full OptiX BVH Patch Traversal

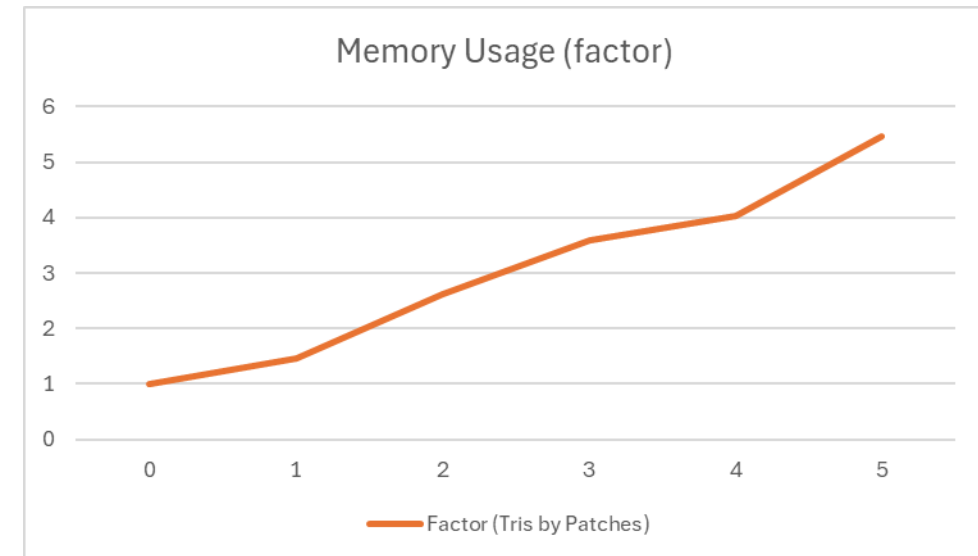
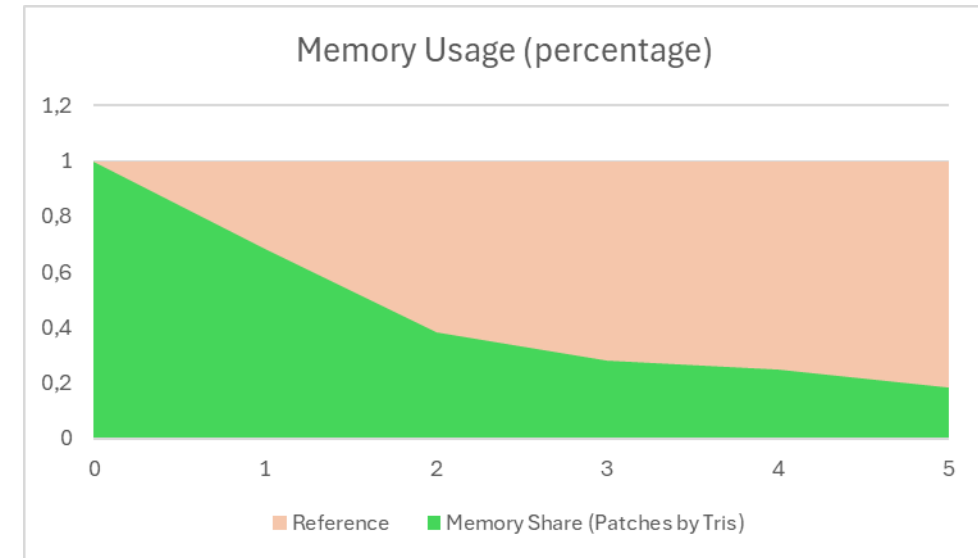
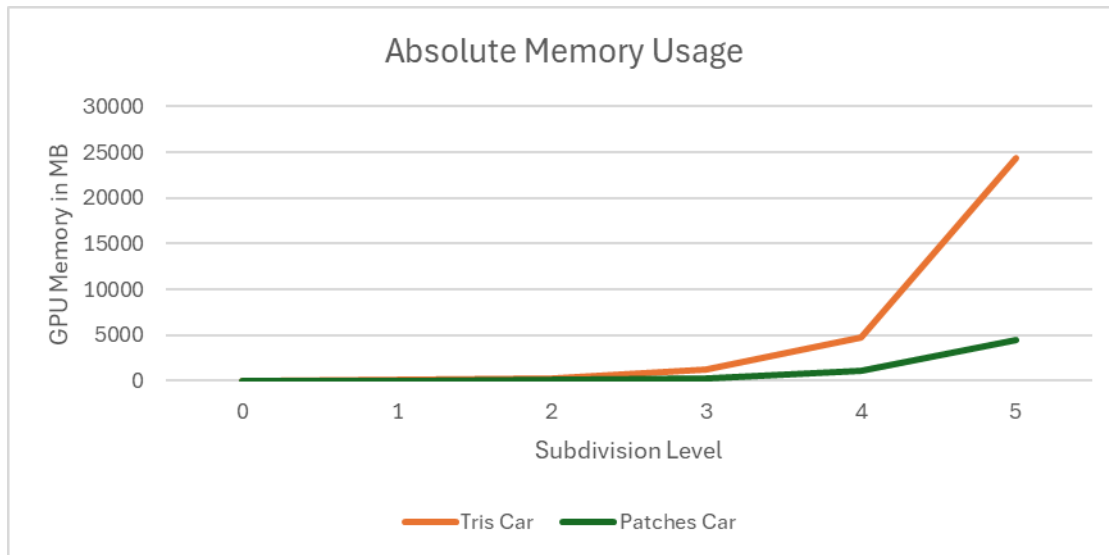
5 Evaluation

Time



5 Evaluation

Memory



6 Outlook

| Displacement – Base Mesh



6 Outlook

| Displacement – Subdivided Mesh



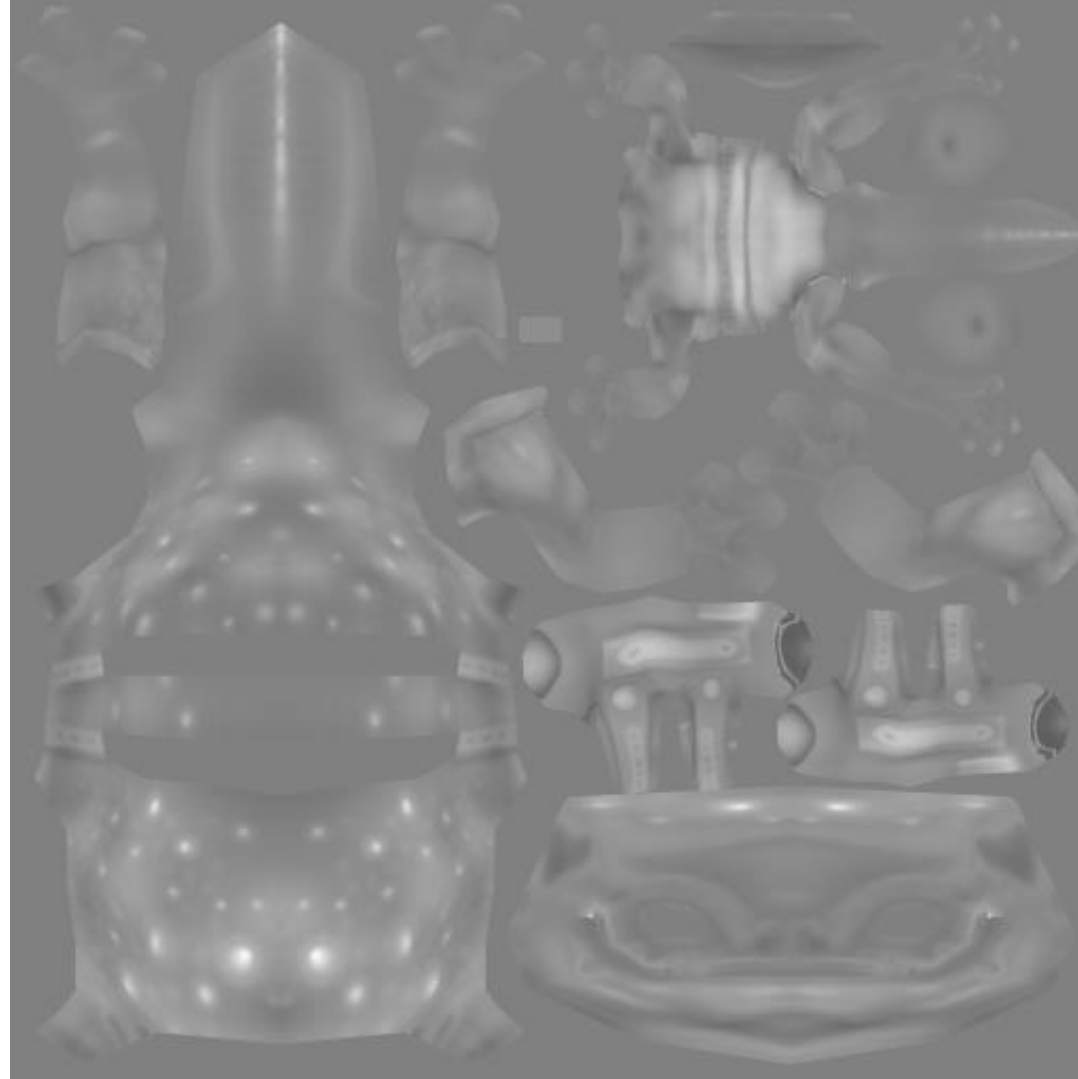
6 Outlook

| Displacement – Subdivided and Displaced Mesh



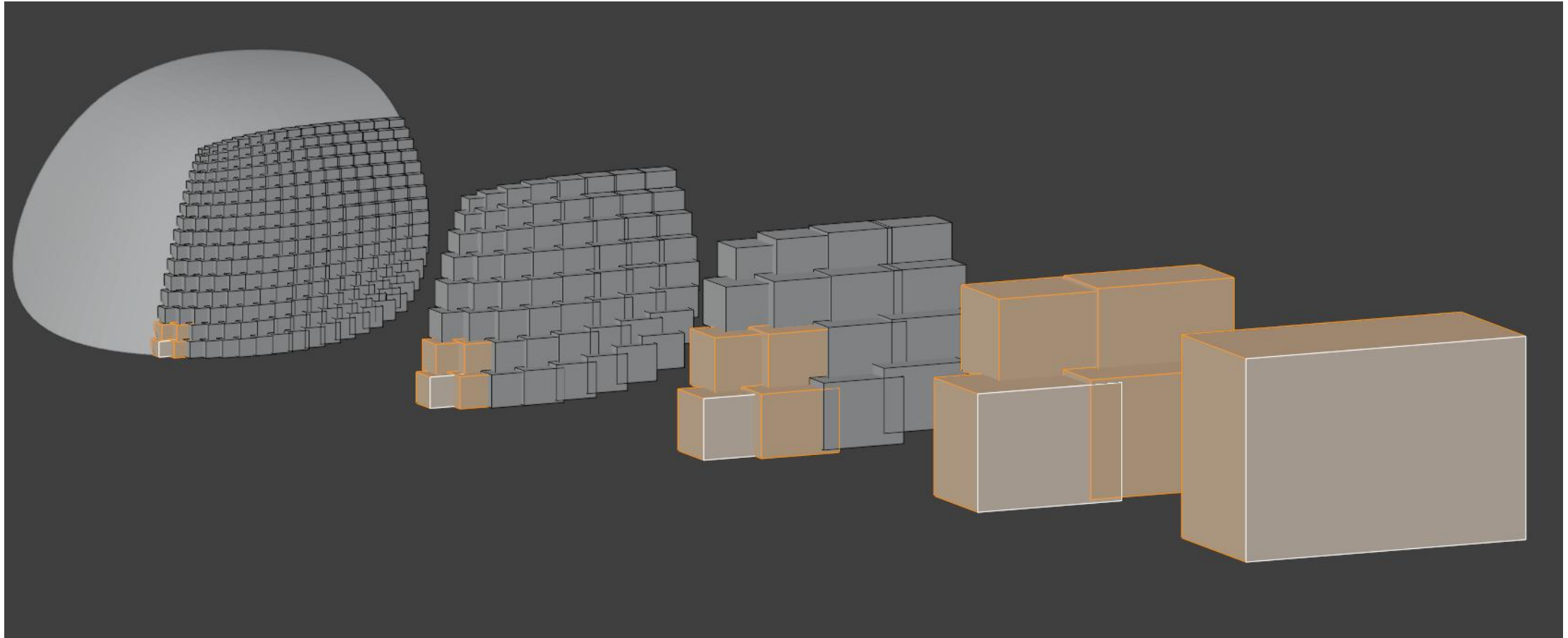
6 Outlook

| Displacement – Displacement Map



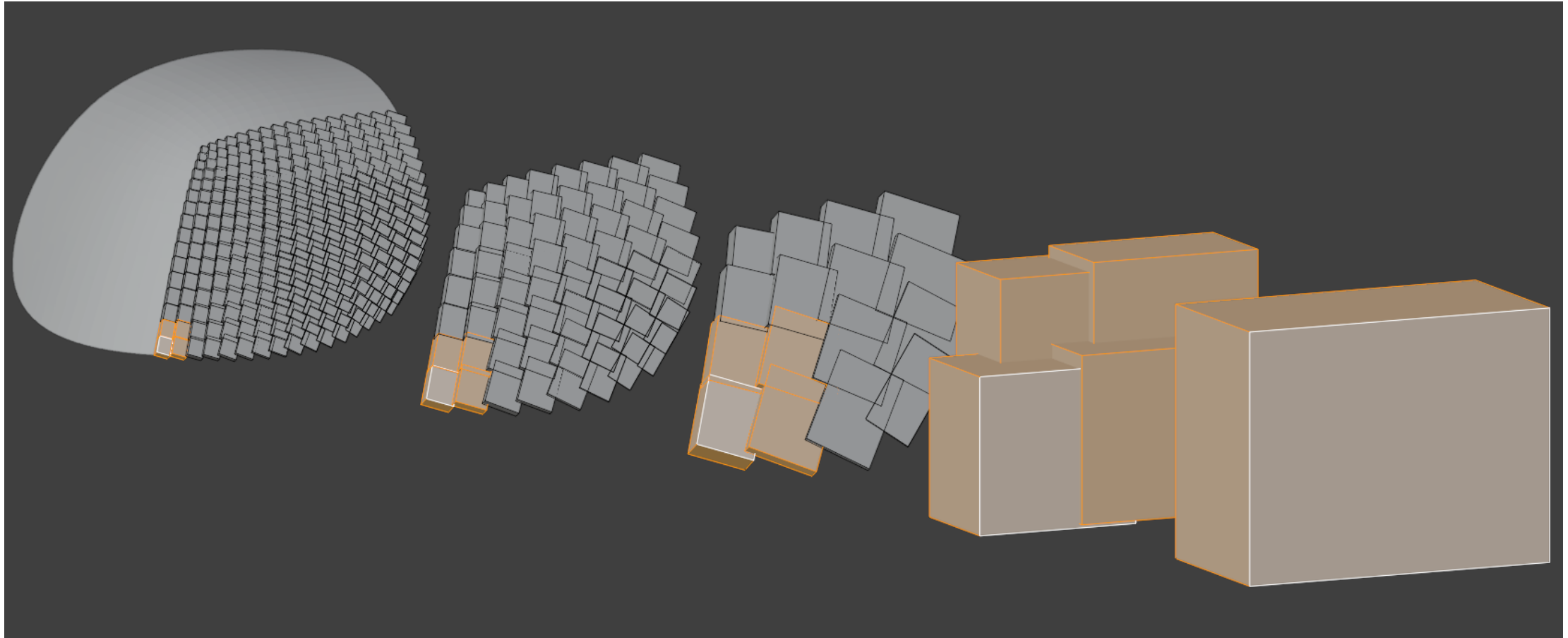
6 Outlook

Compression Scheme – Axis Aligned Bounding Boxes (until now)



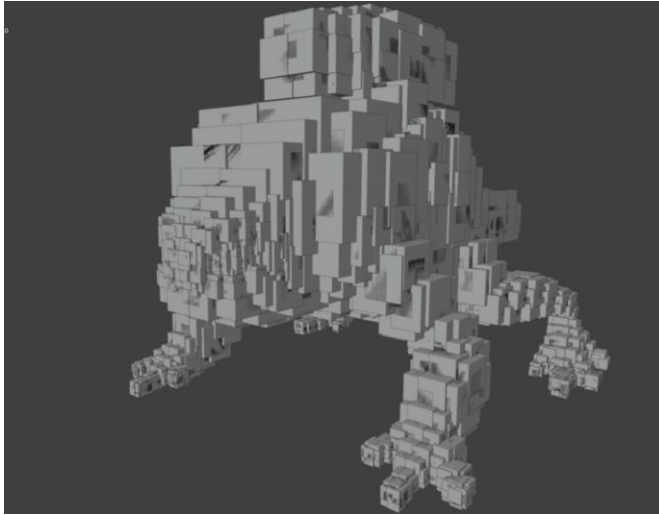
6 Outlook

Compression Scheme – Object Aligned Bounding Boxes

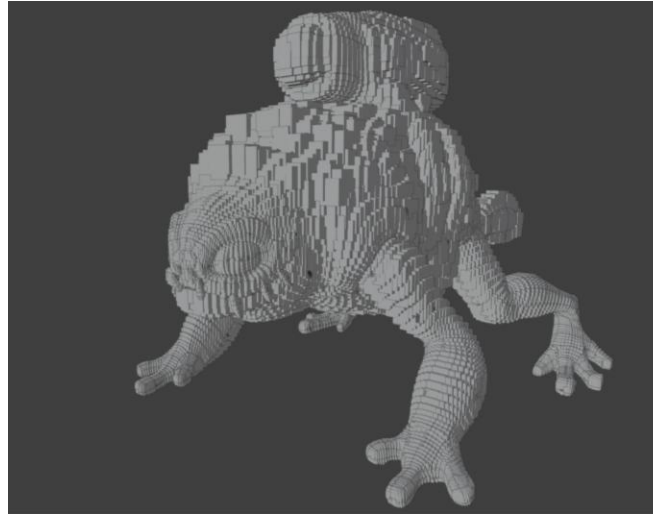


6 Outlook

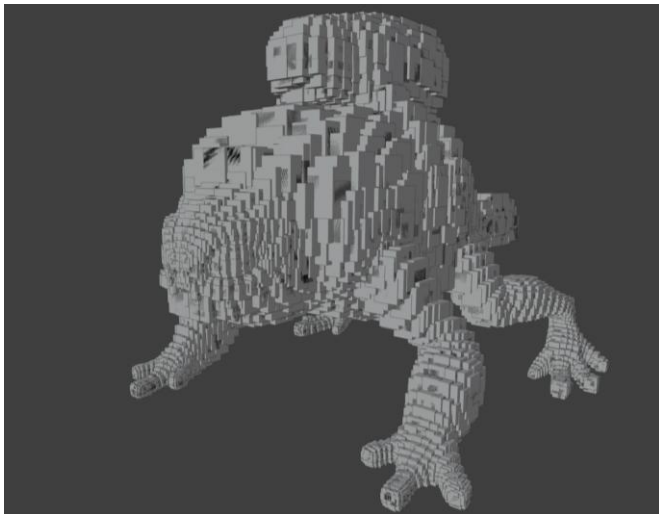
Compression Scheme – Box Approximation



AABBs Level 0



AABBs Level 2



AABBs Level 1



OABBs Level 3



Subdivided Level 3

Further Compression Techniques
(SLM16, LMSS18)

Outtakes

Wrong Textures and Random Displacement

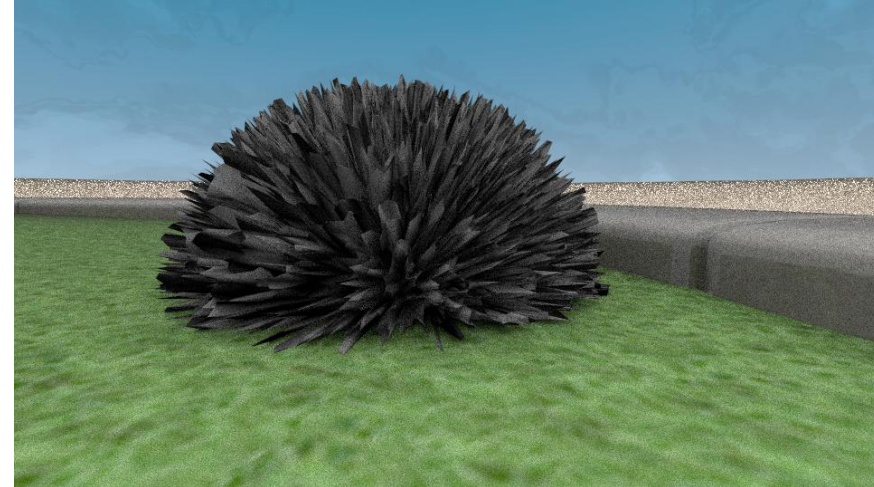
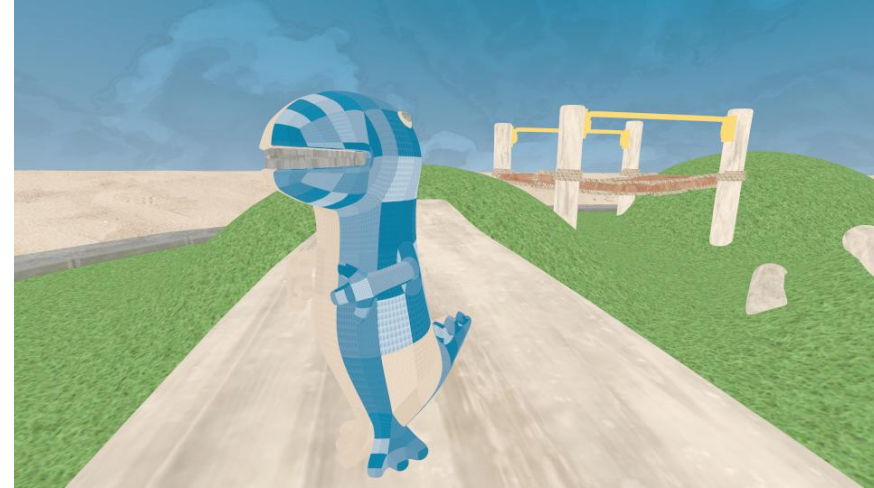
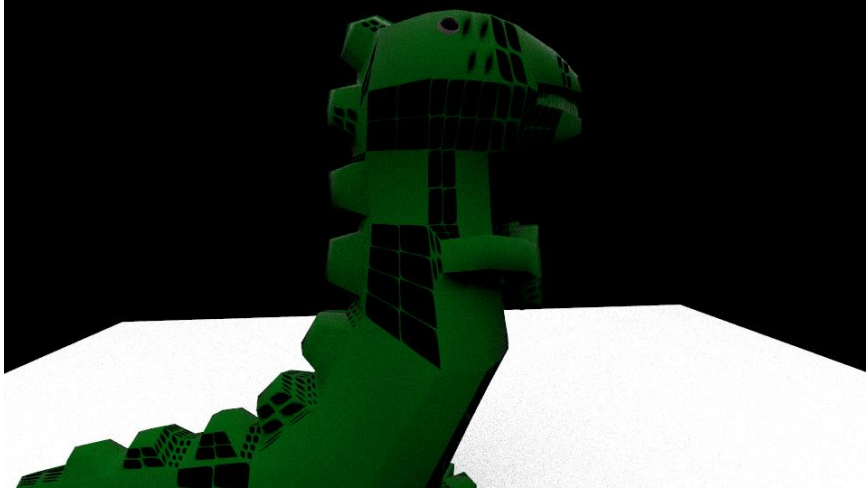


Image Sources

| Links

- NVIDIA OptiX Logo
<https://www.nvidia.com/de-de/gpu-cloud/hpc-visualization-containers/>
- NVIDIA RTX 5090 image
<https://www.nvidia.com/de-de/geforce/graphics-cards/50-series/>